Group Project: Real-World Dynamic Network Application

To deepen your understanding of the major themes in Dynamic Networks, your group will identify a meaningful real-world network problem and develop a software application that demonstrates one or more concepts covered in CECS 427. Each team will consist of **six students**.

Your project should address a meaningful real-world network problem and clearly apply at least one CECS 427 concept. Possible areas include:

- paths, connectivity, and network structure
- social ties, homophily, and community formation
- information diffusion, influence, and cascades
- structural balance and polarization
- strategic behavior, routing, incentives, and market interactions
- network formation, resilience, or changes over time

Project Requirements

1. Software Application

Each team must design, implement, test, and demonstrate a working software application. The application must:

- clearly represent a network using nodes and edges
- implement at least one meaningful CECS 427 algorithm, model, or process
- accept data, parameters, selected nodes, or other user input
- produce a meaningful result, such as a path, visualization, measurement, simulation, comparison, or recommendation
- be tested using several inputs or scenarios

Possible methods include shortest-path search, random-network generation, clustering or neighborhood-overlap analysis, community detection, homophily or structural-balance analysis, information diffusion, and routing equilibrium.

Teams may use any appropriate programming language, framework, or library. A small public, student-created, or simulated dataset is acceptable.

The application may be a web, mobile, or desktop application, simulation, interactive visualization, analytical dashboard, or another approved format. The application is expected to be a functional prototype, not a production-ready system.

2. Source Code and Documentation

Teams may use any appropriate programming language, framework, or library. A small public, student-created, or simulated dataset is acceptable.

The application may be a web, mobile, or desktop application, simulation, interactive visualization, analytical dashboard, or another approved format. The application is expected to be a functional prototype, not a production-ready system.

2. Source Code and Documentation

Submit the complete source code through a team repository or another instructor-approved method. The submission must include:

- all source-code files
- a README describing the project
- installation and execution instructions
- required dependencies and sample data
- instructions for testing the application
- citations for external code, datasets, libraries, tutorials, and other resources
- a brief description of each member's contributions

The instructor should be able to run the application by following the submitted instructions.

3. Project Report

Submit a two-page report formatted in 12-point font, single-spaced, with 1-inch margins. The page limit excludes the title page and reference list. The report should clearly include:

- **Problem Statement:** Describe the real-world problem and explain why it is meaningful in a network context.
- **Network Perspective:** Define the nodes, edges, and relevant network properties. Explain the connection to CECS 427.
- **Application and Algorithm:** Describe what the application does and explain the network algorithm, model, or process implemented.
- **Implementation:** Identify the programming languages, frameworks, libraries, and data used.
- **Testing and Results:** Present several examples or test scenarios and explain the results.
- **Evaluation:** Discuss the application's strengths, limitations, and possible improvements.
- **Team Contributions:** Briefly identify each member's principal contributions.
- **References:** Cite all external sources and materials used.

4. Class Presentation and Demonstration

Each team will deliver a 15–18 minute presentation, followed by questions from the class. Your presentation should:

- introduce the problem clearly

presentation should:

- introduce the problem clearly
- explain the network representation
- connect the project to CECS 427 concepts
- explain the application and its main algorithm
- demonstrate the working application
- present example results
- discuss limitations and possible improvements
- ensure that all six team members contribute meaningfully

Each team member should be prepared to answer questions about the application, network concepts, and code.

5. Academic Integrity and Attribution

All external materials—including code, datasets, libraries, tutorials, articles, images, and existing projects—must be properly credited. Teams must clearly distinguish their own contributions from external work.

Students are responsible for understanding and explaining all submitted code, including code created with assistance from generative AI. Copying an existing application or making only minor changes to it will not satisfy the project requirements.

Example Inspiration

One possible source of inspiration is Rani-Codes' **Six Degrees of Wikipedia** project. The project represents Wikipedia links as a directed network and uses breadth-first search to find and visualize paths between people.

This example is particularly relevant to the CECS 427 topics of graph representation, directed networks, breadth-first search, shortest paths, and network visualization.

Teams are not expected to reproduce this project or match its scale. It is provided only as an example of how a network concept and algorithm can be developed into a user-facing application. Copying the project or making only minor modifications will not satisfy the project requirements.

External source credit: Rani-Codes, Six Degrees of Wikipedia, GitHub (https://github.com/Rani-Codes/sixth_degree)

For inspiration, you may consult relevant **academic sources** such as the journal Social Networks (https://www.sciencedirect.com/journal/social-networks/issues) .